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1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Task 1:

```
lasya_grandhi@DESKTOP-7Q7: x + v
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ nano question1.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ gcc question1.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ ./a.out
Enter a Linux command: ls

--- Parent Process ---
Parent PID: 583

--- Child Process ---
Child PID : 584
\'
a.out      arithmetic.c      execs.c      p2.c.save    process2.c    question2.c    question6.c
addition.c arithmetic_switch  execworking.c pro          program1.c    question3.c    system
arithmetic dir3            file.c      process1     question1.c  question4.c
p2.c      question1.c.save  question5.c

Child process completed.
```



lasya_grandhi@DESKTOP-7Q7



GNU nano 8.7.1

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>
int main() {
char command[100];
pid_t pid;
int status;
printf("Enter a Linux command: ");
scanf("%s", command);    pid = fork();
if (pid < 0) {
printf("Fork failed!\n");
exit(1);
}else if (pid == 0) {
printf("\n--- Child Process ---\n");
printf("Child PID : %d\n", getpid());
execlp(command, command, NULL);
perror("Command execution failed!\n");
exit(1);
}else {
printf("\n--- Parent Process ---\n");
printf("Parent PID: %d\n", getpid());
wait(NULL);
printf("\nChild process completed.\n");
}
return 0;
}
```

Task 2:

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ uname  
Linux
```

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ lscpu  
Architecture:          x86_64  
CPU op-mode(s):        32-bit, 64-bit  
Address sizes:          39 bits physical, 48 bits virtual  
Byte Order:             Little Endian  
CPU(s):                 8  
On-line CPU(s) list:   0-7  
Vendor ID:              GenuineIntel  
Model name:             11th Gen Intel(R) Core(TM) i7-1185G7 @ 3.00GHz  
CPU family:             6  
Model:                  140  
Thread(s) per core:     2  
Core(s) per socket:     4  
Socket(s):              1  
Stepping:               1  
BogoMIPS:               3609.59  
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr ss  
e sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc arch_perfmon rep_good nopl xtopology  
tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pdcm pcid s  
se4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor la  
hf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsb  
ase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid avx512f avx512dq rdseed adx smap avx512ifma  
clflushopt clwb avx512cd sha_ni avx512bw avx512vl xsaveopt xsavec xgetbv1 xsaves vnmi avx5  
12vbmi umip avx512_vbmi2 gfni vaes vpclmulqdq avx512_vnni avx512_bitalg avx512_vpopcntdq rd  
pid movdiri movdir64b fsrm avx512_vp2intersect md_clear ibt flush_l1d arch_capabilities  
  
Virtualization features:  
Virtualization:         VT-x  
Hypervisor vendor:      Microsoft  
Virtualization type:    full  
Caches (sum of all):  
L1d:                    192 KiB (4 instances)  
L1i:                    128 KiB (4 instances)  
L2:                     5 MiB (4 instances)
```

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ lscpu  
Virtualization:         VT-x  
Hypervisor vendor:      Microsoft  
Virtualization type:    full  
Caches (sum of all):  
L1d:                    192 KiB (4 instances)  
L1i:                    128 KiB (4 instances)  
L2:                     5 MiB (4 instances)  
L3:                    12 MiB (1 instance)  
NUMA:  
NUMA node(s):           1  
NUMA node0 CPU(s):      0-7  
Vulnerabilities:  
Gather data sampling:    Not affected  
Ghostwrite:             Not affected  
Indirect target selection: Mitigation; Aligned branch/return thunks  
Itlb multihit:          Not affected  
L1tf:                   Not affected  
Mds:                    Not affected  
Meltdown:               Not affected  
Mmio stale data:        Not affected  
Old microcode:          Not affected  
Reg file data sampling:  Not affected  
Retbleed:               Mitigation; Enhanced IBRS  
Spec rstack overflow:   Not affected  
Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl  
Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization  
Spectre v2:             Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI SW lo  
op, KVM SW loop  
Srbds:                  Not affected  
Tsa:                    Not affected  
Tsx async abort:        Not affected  
Vmscape:                Not affected
```

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ lsblk  
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS  
sda   8:0    0 356.9M  1 disk  
sdb   8:16   0 159.4M  1 disk  
sdc   8:32   0     4G   0 disk [SWAP]  
sdd   8:48   0     1T   0 disk /mnt/wslg/distro
```

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ ps
```

```
PID TTY          TIME CMD
684 pts/0        00:00:00 bash
774 pts/0        00:00:00 ps
```

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ top
```

```
top - 05:31:48 up 11 min, 1 user, load average: 0.05, 0.08, 0.08
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 15857.6 total, 15302.2 free,  590.1 used,  162.0 buff/cache
MiB Swap: 4096.0 total, 4096.0 free,  0.0 used, 15267.5 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24320	15444	11660	S	0.0	0.1	0:00.75	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
6	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
46	root	19	-1	50400	16764	15484	S	0.0	0.1	0:00.15	systemd-journal
78	systemd+	20	0	22416	14532	12032	S	0.0	0.1	0:00.06	systemd-resolve
84	root	20	0	35412	12460	9320	S	0.0	0.1	0:00.36	systemd-udev
104	root	20	0	2888	2012	1884	S	0.0	0.0	0:00.05	chronyd-starter
105	root	20	0	4472	3220	2972	S	0.0	0.0	0:00.00	cron
106	message+	20	0	8820	5468	4792	S	0.0	0.0	0:00.09	dbus-daemon
146	root	20	0	44096	29828	14688	S	0.0	0.2	0:00.34	networkd-dispat
152	root	20	0	18436	9480	8236	S	0.0	0.1	0:00.08	systemd-logind
181	syslog	20	0	220548	5780	4560	S	0.0	0.0	0:00.10	rsyslogd
194	root	20	0	5492	2828	2644	S	0.0	0.0	0:00.00	agetty
225	_chrony	20	0	20424	6392	5616	S	0.0	0.0	0:00.02	chronyd
236	_chrony	20	0	12096	2464	1824	S	0.0	0.0	0:00.00	chronyd
262	root	20	0	123460	32868	18256	S	0.0	0.2	0:00.14	unattended-upgr
319	root	20	0	8644	5676	4836	S	0.0	0.0	0:00.03	login
364	lasya_g+	20	0	22748	13512	11060	S	0.0	0.1	0:00.15	systemd
366	lasya_g+	20	0	22920	4116	2112	S	0.0	0.0	0:00.00	(sd-pam)
397	lasya_g+	20	0	6136	5480	3928	S	0.0	0.0	0:00.04	bash
679	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
681	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.02	Relay(684)
684	lasya_g+	20	0	6268	5664	3992	S	0.0	0.0	0:00.05	bash
791	lasya_g+	20	0	8156	6052	3960	R	0.0	0.0	0:00.00	top